

ECE 4375: Automatic Control Systems

Homework Assignment 1

Differential Equations, the Laplace Domain, Transfer Functions, and Block Diagrams

Name: _____

Date: _____

Instructions: Show all work for full credit. Box your final numerical or symbolic answers where applicable. Unless otherwise stated, assume all initial conditions are zero, the input is $u(t) \leftrightarrow U(s)$, and the output is $y(t) \leftrightarrow Y(s)$.

- 1. First-order differential equation to transfer function.** A system is described by the time-domain differential equation

$$\dot{y}(t) + 4y(t) = 3u(t), \quad y(0) = 0.$$

- (a) Take the Laplace transform of both sides and write the resulting algebraic equation in terms of $Y(s)$ and $U(s)$.
- (b) Solve for the transfer function $T(s) = Y(s)/U(s)$.
- (c) Identify the system's pole and comment on whether the system is stable.

- 2. Second-order differential equation with input derivative.** A system is described by

$$\ddot{y}(t) + 6\dot{y}(t) + 8y(t) = 2\dot{u}(t) + 5u(t), \quad y(0) = \dot{y}(0) = 0.$$

- (a) Take the Laplace transform and write the algebraic equation in terms of $Y(s)$ and $U(s)$.
- (b) Solve for the transfer function $T(s) = Y(s)/U(s)$, written in proper form (numerator and denominator in descending powers of s).
- (c) Factor the denominator and identify the poles of the system.

- 3. Transfer function to block diagram.** Consider the transfer function found for the system in Problem 2,

$$\frac{Y(s)}{U(s)} = \frac{2s + 5}{s^2 + 6s + 8}.$$

- (a) Draw a single-block diagram representation of this system, clearly labeling the input $U(s)$, the output $Y(s)$, and the transfer function block $T(s)$.
- (b) Factor the denominator into $(s + p_1)(s + p_2)$ and identify p_1 and p_2 .
- (c) Suppose this transfer function represents two subsystems in a signal path. Explain (in words) what condition on their individual transfer functions $G_1(s)$ and $G_2(s)$ would allow them to be combined into a single cascaded block equal to $T(s)$.

- 4. Reverse direction: transfer function to differential equation.** A system has the transfer function

$$\frac{Y(s)}{U(s)} = \frac{s + 7}{s^2 + 5s + 6}.$$

- (a) Cross-multiply to write the Laplace-domain algebraic equation relating $Y(s)$ and $U(s)$.
- (b) Take the inverse Laplace transform (term by term, using the derivative property) to recover the time-domain differential equation relating $y(t)$ and $u(t)$.
- (c) Factor the denominator to identify the poles, and state whether the system is stable.

5. Third-order system: differential equation, transfer function, and block diagram. A system is described by

$$\ddot{y}(t) + 2\ddot{y}(t) + 5\dot{y}(t) + 10y(t) = 3\ddot{u}(t) + u(t), \quad y(0) = \dot{y}(0) = \ddot{y}(0) = 0.$$

- (a) Take the Laplace transform and solve for the transfer function $T(s) = Y(s)/U(s)$ in proper form.
- (b) Factor the denominator. (*Hint*: one factor is a real first-order term, and the other is an irreducible quadratic.)
- (c) Draw the single-block diagram for this system, labeling $U(s)$, $Y(s)$, and $T(s)$.

6. Mass–spring–damper system: full derivation and block-diagram construction. A mass–spring–damper system with mass M , damping coefficient D , and spring constant K is driven by an external force input $u(t)$, producing displacement output $y(t)$. Newton's second law gives

$$M\ddot{y}(t) + D\dot{y}(t) + Ky(t) = u(t), \quad y(0) = \dot{y}(0) = 0.$$

- (a) Take the Laplace transform and solve for the open-loop transfer function $T(s) = Y(s)/U(s)$ in terms of M , D , and K .
- (b) Let $M = 2$ kg, $D = 8$ N·s/m, and $K = 6$ N/m. Substitute these values into $T(s)$ and simplify to a numerical transfer function.
- (c) Factor the denominator of the numerical transfer function from part (b) and identify the poles.
- (d) Draw a block diagram for the numerical system from part (b) using only integrator blocks ($1/s$) and static gain blocks, arranged with summation junctions, so that the diagram relates $U(s)$ to $Y(s)$ without a single lumped $T(s)$ block. (*Hint*: solve the differential equation for the highest-order derivative of $y(t)$ first, then work backward through successive integrators.)